Which evidence deserves more weight?

Compare how evidence was gathered before deciding how much it supports a claim.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None; all research terms are explained and no calculations are required.

Students will be able to

- Compare evidence using participants, comparison groups, and outcome measures.
- Explain why a method strengthens support without treating it as a guarantee.
- State a relevant limit on transferring a result to another setting.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; no student devices or calculator required.
- Instructor guide with model answers, kept separate from the handout.
- Read the vocabulary and the supplied report descriptions.
- Print the three-page handout or display the activities in turn.
- Explain that all reports and figures are invented. Learners should judge the described evidence, without inventing missing study details.

Teaching sequence

5 min	**Separate experience from a comparison.** Learners complete the flashcard warm-up. Introduce testimonial and comparison group, using the model answer to explain why a second group helps.
10 min	**Compare the tutoring reports.** Pairs complete the main activity. Explain random assignment and outcome measure. Ask them to connect every reason to a stated feature, not simply say the report is scientific.
8 min	**Discuss strength and limits.** Compare pair responses with the model. Use both discussion prompts and correct the idea that sample size alone establishes cause or that one favorable study applies everywhere.
7 min	**Assess a new comparison.** Learners complete the spelling exit ticket independently. Check their reasons and limitation against the assessment criteria, then reveal the answer.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / Do flashcards help?

Student task and instructor answer

Explain these words as needed

Testimonial: A person's report of their own experience; it may be sincere without showing what caused the experience.

Comparison group: A group used to judge what happens under a different condition, such as usual support instead of tutoring.

Random assignment: Using chance to allocate participants to groups. It reduces systematic group differences on average; it does not guarantee perfectly identical groups.

Outcome measure: The recorded result used to evaluate a claim, such as passing the same algebra exam.

Scenario

Invented scenario: A tutor claims flashcards improve vocabulary test scores. Report A quotes one learner: 'It really helped me.' Report B compares two groups of 20 learners with similar starting scores, but no random assignment: flashcards versus usual review sheets. Both groups review for the same time and take the same test. All scores are reported; the flashcard group has the higher average.

Student output: Write three sentences: choose a report, explain one strength in how its evidence was gathered, and name one remaining uncertainty.

1. Choose which report gives stronger evidence for the tutor's claim.

2. Explain how either its comparison group or common test helps assess the claim.

3. State one reason the description does not establish the method's effect with certainty.

Student success check: Can you explain your choice without treating a sincere testimonial as proof or treating a comparison as a guarantee?

Model answer

Report B gives stronger evidence. Its common test compares the claimed outcome after two review methods with equal review time. Because learners were not randomly assigned, differences such as motivation could still partly explain the higher average; the scenario does not tell us whether that happened.

Why this meets the task: The answer identifies a relevant design advantage and distinguishes an untested possible explanation from an established fact. A testimonial may describe a real experience but does not supply the same comparison.

Main activity / Two reports about algebra tutoring

Student task and instructor answer

Scenario

Invented scenario: A tutoring program claims it raises algebra pass rates. Report A quotes one volunteer saying tutoring helped. Report B follows 200 students randomly assigned to receive tutoring or usual support, uses the same exam, and reports all results: 70 of 100 tutored students pass, versus 55 of 100 others. Both reports are accurately described; use only these facts.

Student output: Write a choice, three reasons about the method, and one question about applying the result to another school.

1. Choose the stronger evidence for the program's claim.
2. Give three reasons using the number of students, the comparison groups, and the outcome measure.
3. State one limit: what would you still want to know before assuming the result applies to another school?

Student success check: Explain the evidence's method rather than treating a confident testimonial or a large number alone as decisive.

Model answer

Report B is stronger: it includes 200 students rather than one selected volunteer, random assignment supplies a fairer comparison, and a common exam measures the stated outcome. I would check whether another school's students and tutoring program are similar before transferring the conclusion.

Why this meets the task: The answer connects three stated design features to the claim and keeps a local result separate from a universal promise. Random assignment makes it easier to judge whether tutoring caused the difference, but the description does not justify certainty about every student or setting.

Exit ticket / Testing a spelling program

Student task and instructor answer

Scenario

Invented scenario: A school claims its new spelling program improves spelling test scores. Report A quotes one parent saying the program worked for their child. Report B records 80 pupils randomly assigned to the new program or usual spelling lessons for six weeks. Both groups receive equal lesson time and take the same final test; all results are reported. The new-program group's average is 16 out of 20, compared with 13 out of 20 for the usual-lessons group. No information about other schools is supplied.

Student output: Write the stronger report, three reasons tied to its method, and one question to ask before using the result in another school.

1. Choose the stronger report for the stated claim.
2. Explain how the participants, comparison groups, and common outcome measure affect the strength of the evidence.
3. State a relevant detail you would check before assuming another school's pupils would benefit similarly.

Student success check: Have you explained the method, kept your conclusion about the reported comparison, and avoided inventing a result for another school?

Model answer

Report B is stronger. It includes 80 pupils rather than a selected parent's single experience; chance assignment and equal lesson time make the new program and usual lessons a fairer comparison; and the same final test measures the claimed outcome. Before applying the result elsewhere, I would check whether pupils' starting spelling skills and the way the program is delivered are similar. The reported higher average does not show that every pupil improved or that every school would get the same result.

Why this meets the task: The answer connects the described methods to the claim, identifies a relevant transfer question, and states limits without dismissing the evidence or pretending the limits have already been resolved.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- The stronger report is chosen with three reasons tied to stated methods.
- The explanation goes beyond sample size and addresses comparison and measurement.
- A relevant question about transferring the result is supplied without claiming certainty, universal benefit, or invented facts.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Why is '200 students' not enough by itself to make the tutoring evidence strong?

Listen for: A large group can still be badly selected or measured. The comparison, random assignment, common exam, and complete reporting also matter.

Does stronger evidence make the testimonial worthless or prove that every tutored student benefits?

Listen for: No. A testimonial can suggest questions or report experience, but is weaker for estimating a program's effect. Group results also do not establish an identical effect on every individual.

Common confusions

A bigger sample automatically proves cause. Ask how the groups were formed and what was measured. More observations do not repair every weakness in a method.

Random assignment means participants were randomly chosen from everyone. Assignment concerns allocation to groups after participation; recruitment concerns who enters the study. The scenario does not describe recruitment.

A higher group average means every learner improved. An average summarizes a group. Individuals may have different outcomes; do not infer all individual changes from the mean.

Adapt the lesson

More support: Use three headings for reasons: Who was included? What was compared? What was measured? Read each report aloud and have pairs locate the matching facts before explaining them.

More challenge: Ask: 'If both spelling groups used different final tests, what would become harder to interpret?' Expected response: the score difference could reflect test difficulty rather than the teaching method; scores would no longer provide the stated common measure.

Next practice

https://xixteen.com/learners/?practice=evidence_quality#workbench